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Supply chain vertical common ownership and cost of loans

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ABSTRACT

This study explores how vertical common ownership reduces supply chain risk and influences creditors' decisions, focusing on the cost of loans. Findings reveal that creditors view such firms as less likely to default, evidenced by lower loan spreads. This effect is stronger for suppliers with more relationship-specific investments, weaker bargaining power, higher information asymmetry, and long-term common institutional ownership. Results remain robust through quasi-natural experiments from financial institutions' M&As, supplier-customer pair analyses, etc. We also compare vertical common ownership with the horizontal one and demonstrate its incremental contributions. Moreover, the fraction of customer sales and the institutional investors' share ownership likely influence the likelihood of vertical common ownership, with our results consistently holding under Heckman analysis. Overall, the results suggest that vertical common ownership enhances coordination and monitoring, reducing risks and creditor risk premiums. The findings may offer valuable insights into managing supply chain risks and understanding their financial implications.

1. Introduction

Supplier-customer relations are pivotal in the real sector, especially when transactions involve incomplete contracts and relationship-specific investments, leading to the hold-up problem. This issue, as noted by Grossman and Hart (1986), results in potential exploitation and underinvestment. The problem intensifies under the current complex global scenario, characterized by disrupted supply chains and rising protectionism. An illustrative case is the U.S. port crisis, where container pile-ups due to unclaimed cargo have spiked cargo costs, negatively impacting suppliers' investments. Vertical integration could resolve this, but its feasibility is limited by merger risks, high costs, and anti-monopoly laws. Therefore, it is crucial to explore alternative mechanisms that enhance cooperation and reduce opportunism and hold-up risks in supply chains.

Traditional economic theory posits that firms are self-interested, aiming to maximize their value (Smith, 1937). This assumes that major shareholders do not have significant stakes in competing firms. However, recent studies suggest that institutional investors, who aim to maximize their entire portfolio's value, can coordinate behaviors among commonly owned firms. Harford et al. (2011) found that most institutional investors in S&P 500 companies prioritize overall portfolio value over individual firm value. Similarly, Park

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et al. (2019) observed improved performance and increased voluntary information disclosure among companies with common institutional ownership. These insights indicate a shift towards a more cooperative approach in company operations. Given that the primary effect of common ownership is fostering collaboration compared to normal institutional investors, the supply chain is an ideal venue to examine such a topic, as the cooperation between suppliers and customers is especially important. However, most existing literature focuses on horizontal common ownership among competing peers, while less attention has been paid to vertical partners. There are just three notable exceptions. Freeman (2023) and Chen (2021) find that common ownership between a supplier and its customer helps strengthen their relationship and facilitates the supplier firm to invest more in innovation. Based on social network theory, Cheung et al. (2020) find that vertical common ownership increases suppliers' operating and market performance.

In this study, we delve into the effects of vertical common ownership on supply chain hold-up issues and its subsequent influence on stakeholder perceptions, with a specific emphasis on resolving agency conflicts between companies and their creditors. Corporate loans are a predominant source of financing, comprising a significant portion of U.S. corporate debts. By considering creditors as primary stakeholders, we aim to understand how vertical common ownership influences the dynamics in a crucially important creditor relationship, offering a clearer understanding of the broader financial implications of these ownership structures in corporate finance. Our core argument is that, vertical common ownership can i) coordinate the supply chain partners, to maximize their total investment portfolio value; and ii) effectively monitor corporate behaviors, and lower adverse selection and moral hazard issues. Such an influence can affect not only the supplier or customer firm itself, but also the attitude and decisions of outside corporate stakeholders. Thus, as an important outside stakeholder, creditors will perceive vertical common ownership as a positive signal, and further lower their cost of loans.

Noting that we study the role of vertical common ownership on the supplier side. This is mainly because we develop the hypothesis based on hold-up problems and relationship-specific investments. Generally, the relationship-specific investment is provided by the supplier firm to a specific customer firm, after which the supplier is more vulnerable to the customer's rent extraction (Grossman and Hart, 1986). Therefore, we argue that vertical common ownership is more beneficial for supplier firms considering their weaker position in the relationship, rather than customers. In addition, we test the main hypotheses based on the supplier-customer dyad level in the robustness check section. However, the negative relationship may not be held according to the collusion hypothesis and private information sharing hypothesis, which remains an empirical issue.

Using a sample of 7005 facility observations from the year 1986 to 2017, we empirically test the impact of vertical common ownership and the cost of loans. In line with our prediction, there is a negative association between the common ownership of a supplier with its customers and the cost of loans. In robustness analysis, we find similar results in other loan pricing variables (i.e., longer loan maturities, greater loan size, number of lenders, and lower total cost of borrowing). Further cross-sectional tests are in accordance with the coordination and monitoring channels. The results are robust under a battery of robustness checks. We also argue that the fraction of customer sales and the institutional investors' share ownership are likely to influence the likelihood of vertical common ownership. Our results consistently hold under the Heckman analysis.

This study makes the following potential contributions. First, from the supply chain perspective, this study provides new evidence on common ownership literature. As the primary role of common ownership is to foster collaboration, the supply chain is an ideal venue to examine such a topic, though less attention has been paid to this area. Extant studies mostly focus on the common institutional ownership of industry peers (i.e., horizontal common ownership) and find mixed evidence on whether such ownership improves peer cooperation. Most of the prior literature has found a positive relationship between common ownership and cooperation among industry peers. In contrast, Lewellen and Lowry (2021) document that this type of relationship is mainly driven by other factors, such as the 2008 financial crisis. There are also controversies such as the anti-competitiveness effect, limited attention hypothesis, collusion hypothesis, etc. The coordination role of vertical common ownership is essentially different from that of horizontal one; whereas the former improves total utility, the latter is welfare-reducing due to its anti-competitiveness. In this paper, we examine the creditor's perceptions of common ownership between supply chain partners (rather than competing peers within an industry). Empirical results validate the positive role of vertical common ownership in coordinating and monitoring supply chain partners. Results still hold when controlling for horizontal common ownership and institutional shareholdings, which indicates the incremental effect of vertical common ownership and complements prior studies.

Second, we extend the literature on how supply chain relations affect creditors' decision-making. Existing studies document that corporate internal governance and external monitoring can affect the financial decisions of creditors (e.g., Álvarez-Botas and González, 2021; Campello et al., 2011; Francis et al., 2017; Valta, 2012).[1] However, only few of these studies have ever considered the supply chain relationship, which find that durable supply chain relationships (Cen et al., 2016; Croci et al., 2021) and lower customer concentration (Campello and Gao, 2017) lead to lower cost of loans. As we know, this study is the first to connect supply chain vertical common ownership to the cost of loans. Through the coordination and monitoring mechanisms, vertical common ownership reduces creditors' perceptions of supply chain risk significantly, which adds to the existing literature on how supply chain relations affect financial decisions made by creditors.

Lastly, the results of this study potentially mitigate the apprehensions of regulators and policymakers about common ownership. While some earlier research posits that common ownership can oversee corporate conduct, there remains a segment of scholarship that challenges these assertions (Azar et al., 2018; Dennis et al., 2022; Lewellen and Lowry, 2021) or highlights the detrimental effects of

[1] Existing studies have explored how various internal corporate factors like financial reporting (Ertugrul et al., 2017), firm value (Campello et al., 2011), and governance (Fields et al., 2012), along with external elements such as financial intermediaries (Francis et al., 2017), market conditions (Valta, 2012), and institutional frameworks (Álvarez-Botas and González, 2021), significantly impact cost of loans.

common ownership. Given that competition is conducive to societal welfare and the generation of value, the anti-competitive nature associated with (horizontal) common ownership could be detrimental. For instance, common ownership might result in lower executive pay-performance sensitivity (Antón et al., 2021), greater value-destroying M&As, and lower stock returns (Kempf et al., 2017). Our study presents evidence that vertical common ownership can facilitate coordination among supply chain partners and mitigate hold-up issues, thereby boosting stakeholder confidence and leading to a reduction in loan costs. These findings suggest that common ownership can have a positive impact on total welfare, which should assuage the concerns of regulators and policymakers.

2. Literature and hypothesis

2.1. Prior studies on common ownership

Common ownership occurs when an institutional investor holds shares in multiple firms concurrently. This ownership structure is characterized by the investors' aim to maximize the overall value of their combined holdings, rather than focusing on individual firm values (Hansen and Lott, 1996; Gilje et al., 2020). Consequently, common investors tend to coordinate the behavior of the commonly owned firms. Most studies on common ownership focus on commonly owned industry peers and find mixed evidence. He and Huang (2017) observe that firms with commonly owned industry peers have better operating performance, consistent with the coordination argument. Azar et al. (2018) document that common ownership reduces price competition and allows airlines to charge higher prices. Ni and Yin (2023) find that common ownership decreases a firm's cost of capital by lowering product market competition. Brooks et al. (2018) present evidence that common ownership not only increases the likelihood of mergers and acquisitions but also improves post-deal performance and decreases the probability of unsuccessful deals. Park et al. (2019) argue that commonly owned firms face lower proprietary costs and are more inclined to disclose information voluntarily, including management earnings forecasts. Conversely, Harford et al. (2011) find that common ownership matters little in firm acquisition decisions, which is difficult to reconcile with the point that common ownership accounts for the lack of shareholder resistance to bad firm behaviors. Lewellen and Lowry (2021) contest the link between common ownership and firm coordination, suggesting that this relationship is not robust and could be influenced by other factors.

Furthermore, previous studies have highlighted the role of common ownership in monitoring and knowledge sharing, attributed to their informational advantage. Chen et al. (2021) find that cross-owners in multiple same-industry firms possess informational benefits, such as broader access to information and industry expertise. Gao et al. (2019) document that this informational advantage under common ownership fosters knowledge exchange among firms, thus bolstering corporate innovation. He et al. (2020) and Ramalingegowda et al. (2021) observed that common ownership reduces firms' earnings management through improved monitoring. Edmans et al. (2019) discover that common ownership enhances corporate governance across different industries through both active engagement and disinvestment strategies, supporting the monitoring hypothesis. Additionally, He et al. (2024) provide evidence that the common ownership between borrower firms and financial banks can alleviate information frictions and decrease the firm's loan spreads, which arise from the information dissemination within the portfolios.

According to existing literature, less attention has been paid to firms with commonly owned vertical partners, where cooperation between suppliers and their customers is a major issue. Some notable exceptions are Freeman (2023) and Chen (2021), both of which find that supplier firms are more willing to undertake innovative activities when they have commonly owned customer firms, which is consistent with institutional investors' ability to play a coordinating role and thus reduce supplier firms' perception of hold-up risk. Based on social network theory, Cheung et al. (2020) document that vertical common ownership increases suppliers' operating and market performance.

2.2. Hypotheses development

Prior financial literature argues that default risk is the key lending risk faced by creditors and is one of the fundamental factors of loan pricing (e.g., Merton, 1974; Demiroglu and James, 2010). Higher default risk may induce higher cost of loans and stricter loan contracts in order to alleviate agency conflicts between the borrowers and lenders. This relationship is underscored by agency theory, which posits that loan covenants can mitigate agency problems between creditors and shareholders (Jensen and Meckling, 2019; Myers, 1977).[2] In this article, we argue that common ownership along the supply chain can lower creditors' perception of corporate default risk and thus reduce loan spreads of supplier companies because of the coordination and monitoring channels.

First, vertical common ownership can coordinate the behaviors of supply chain partners, lower hold-up problems, and promote a stable supply-customer relationship. When institutional investors hold multiple firms, investors aim to maximize their investments' total portfolio value, rather than the value of individual firms (He and Huang, 2017). In this sense, vertical common investors are incentivized to coordinate the behaviors of their portfolio firms (i.e., supplier and customer). Cen et al. (2016) show that long-term supply chain relationships can act as a reputation signal, which helps diminish informational disparities between enterprises and creditors, and lowers corporate loan costs. Campello and Gao (2017) document that companies with fewer hold-up problems may lead to lower default risk and lower cost of loans. Similar findings are shown in Dhaliwal et al. (2016). Building on these studies, we argue

[2] Smith & Warner (1979) document that there is an ideal set of contracts for the specific characteristics of a certain project. For riskier firms, creditors impose greater spreads to compensate for their default risk and construct stricter contracts to mitigate agency problems (Rauh and Sufi, 2010).

that due to the strengthened supply chain relationships created by vertical common ownership, customer firms are less likely to be opportunistic, and suppliers will seldom be held up, which lowers the hold-up risk and default risk perceived by creditors. Thus, we expect a negative relation between vertical common ownership and suppliers' cost of loans.

Second, in addition to their coordination role, vertical common shareholders have the motivation and ability to perform effective monitoring. Institutional investors can effectively monitor corporate behaviors, thus improving the provision of corporate information and quality of financial reporting and reducing information asymmetry (Appel et al., 2016). Such monitoring ability is intensified when institutional investors are block-holders of multiple firms simultaneously. In addition, for institutional investors who are shareholders of suppliers and their customers simultaneously, the externalities of one side's behaviors are internalized by the other partner they hold, affecting the total portfolio values. Consequently, common owners have stronger motivation in monitoring corporate misbehaviors. And the enhanced monitoring may increase corporate information transparency and promote better corporate governance (He et al., 2019). To the extent that the vertical common institutional investors' monitoring increases, creditors perceive companies with lower operation and information risk, and their concerns about adverse selection and moral hazard in loan contracts are alleviated. Therefore, creditors will provide a reduction in loan pricing.

In brief, vertical common ownership provides new information to creditors, helps improve their confidence in supply chain relationships, and mitigates their concerns about managers' opportunistic behaviors, which further lowers the cost of loans.

Nevertheless, the negative influence on the cost of loans may not be significantly observed. First, based on the collusion hypothesis, supply chain partners may collude for their own benefit but harm the external stakeholders. In such a situation, the monitoring effect of common investors may be undermined due to their resource extraction incentives (Raman and Shahrur, 2008). The common investors are given both opportunities and abilities to extract resources for their own interests (Matvos and Ostrovsky, 2008) and expropriate the wealth from creditors. In this sense, creditors may view vertical common ownership negatively and increase loan pricing. Second, according to the private information sharing hypothesis, customers will reduce the demand for the suppliers' public information when they have better access to suppliers' private information (Crawford et al., 2020). Since common institutional investors can promote information sharing (Freeman, 2023), customers know more information through their common owners. In this regard, considering the proprietary costs and disclosing costs, suppliers do not need to disclose such information, and the public information on which the creditors can rely becomes less. Thus, creditors may compensate for the higher cost of loans. The last reason is due to the possible incapability of creditors. It is unclear whether creditors are able to observe and process the vertical common ownership information effectively and whether or not they regard it as a pricing factor.

Therefore, the baseline hypothesis can be stated as a null hypothesis:

Hypothesis: Others being equal, there is no relationship between vertical common ownership with major customers and a supplier's cost of loans.

3. Research design

3.1. Measure of vertical common ownership

We construct four variables to capture vertical common ownership (He and Huang, 2017; He et al., 2019). The first independent variable, *CO_Firms*, is the natural log of one plus the number of the firm's customers that are co-owned with the supplier firm. The second independent variable, *CO_Firms_MV*, is the natural log of one plus the market value of the firm's customers that are co-owned with the supplier firm. The third independent variable is *COwn_#*, which equals the natural log of one plus the number of institutional investors who are shareholders of both the supplier firm and its major customers. The fourth independent variable is *Avg_Investors*, which is the number of vertical common institutional investors scaled by the number of total institutional investors in the firm. We set 5 % as the shares' cutoff for common institutional owners, in order that the investors can exert enough influence on firm decisions. The institutional shareholding data from Thomson Reuters 13F are provided at the quarter level. Since stakeholders may evaluate the company over a period of time, we average the quarter common ownership measurements into the most adjacent fiscal year to which the loan facility belongs.

3.2. Measure of cost of loans

As bank loans remain a predominant source of corporate financing, comprising approximately two-thirds of U.S. corporate debts (e.g., Carey and Nini, 2007; Houston and James, 1996),[3] we use loan spread (*Spread*) as the main measurement of the cost of loans in our empirical analysis. *Spread* is the natural log value of the all-in spread drawn, charged by the bank over LIBOR for the drawn portion of the loan facility, as reported in the LPC DealScan's facility-level data. In our robustness analysis, we also investigate the effects on loan maturity (*Maturity*), loan amount (*LoanSize*), and the number of lenders in a loan facility (*NoLenders*). Additionally, we use total cost of borrowing for complementary analysis (Berg et al., 2016).

[3] As the dominant external financing source, debt financing involves borrowing funds and repaying them with interest. Bank loans are the most prevalent type of corporate debt financing. Houston and James (1996) document that two-thirds of the debts for U.S. companies are bank loans. Carey and Nini (2007) also provide evidence that U.S. private lending operations are predominantly influenced by the position of domestic banks as credit institutions.

3.3. Regression model

Following Campello and Gao (2017) and Cohen et al. (2014), regression model is shown in eq. (1).

$$Spread_{i,k,t} = \alpha + \beta \times CO_{i,t} + \gamma' FirmControls_{i,t} + \delta' LoanControls_{k,t} + Loan\ Purpose\ FE + Loan\ Type\ FE + Year\ FE + Firm\ FE + \varepsilon_{i,k,t} \quad (1)$$

Here, the subscripts *i, k*, and *t* denote firm *i*, loan facility *k*, and year *t*, respectively. The dependent variable, *Spread*, represents the loan spread, which is the most common measurement of the cost of loans.[4] *CO* captures firms' common ownership status. We use four common ownership variables, namely *CO_Firms, CO_Firms_MV, COwn_#,* and *Avg_Investor*.

For control variables, firm-level controls include firm size (*SIZE)*, leverage (*LEV)*, market to book (*MTB)*, *ROA*, *Tangibility*, cash flow volatility (*CFO_VOL)*, Altman (1968) *Z*-Score (*Z_Score*), and bond rating dummy (*Rating)*. In addition, loan characteristics controls include loan maturity (*Maturity)*, loan amount (*LoanSize)*, and performance pricing dummy (*Performance)*. Detailed variable definitions are seen in Appendix A.

Moreover, throughout the empirical analysis, we use loan purpose and loan type fixed effects, year fixed effects, and firm fixed effects to alleviate potential omitted variable concerns. Loan purposes include corporate purposes, working capital, debt repayment, acquisition, backup line for commercial paper, and others. Loan types include term loans, revolvers, and 364-day facilities. The year fixed effects represent the period a loan facility belongs to. Continuous variables are winsorized at 1 % and 99 %.

3.4. Data and sample

Data used in this study are from various sources. We obtain supplier chain data from the Compustat Segment file in WRDS, which provides comprehensive coverage of major buyer-supplier relationships and is widely used in academic research (e.g., Gu et al., 2017; Radhakrishnan et al., 2014). According to the mandate by *the Statement of Financial Accounting Standards No. 14* and *Statement No. 131*, public firms are required to disclose their major customers who account for at least 10 % of total sales.[5,6] Common ownership measures between suppliers and customers are constructed using the Thomson Reuters 13F database. Loan cost data are determined by Thomson Reuters loan pricing corporation (LPC) Dealscan, which comprises a variety of loan features. The financial data of control variables are from Compustat.

As for dataset matching, following Cen et al. (2017) and Cohen and Frazzini (2008), we utilize the WRDS Analytics Tool, which is built on the historical customer piece from Compustat Segment data. This tool employs a fuzzy name-matching algorithm to link customer identifiers across historical CRSP and Compustat company fields, ensuring accurate integration of datasets. Each match is manually verified to enhance the reliability of our common ownership measures. Additionally, the records are calibrated and complemented with publicly available data and contributions from researchers.[7]

We start constructing the sample from supply-customer relationship data, which includes 30,356 pairs in a total of 115,877 observations ranging from 1976 to 2017. Then we drop pairs without disclosing the date or sales, and match the remaining data with the 13F database. The institutional shareholding data in 13F is provided at the quarter level, and we obtain 86,494 quarter observations for 24,244 pairs. We further aggregate the total quarterly supplier-customer pairs into supplier-level data, and average them within a fiscal year.

Later on, we calculate the cost of loans through Dealscan, and merge the loan facility with Compustat, CRSP, and common ownership data using the linktable from Dealscan. For financial variables, book value of total assets (or equity) should be greater than zero, and all the financial and loan characteristics variables are not missing. Finally, the baseline sample comprises 7005 facility (or tranche) observations from the year 1986 to 2017.

4. Results

4.1. Statistics

Table 1 presents the summary statistics for main variables employed in this study. Consistent with Graham et al. (2008) and

[4] The subscript *t* in *Spread* specifically indicates the year of loan initiation. Essentially, $Spread_{i,}$ is equivalent to $Spread_{i,k,t}$; the latter notation explicitly incorporates the time dimension into the model.

[5] Our analysis reveals that 76.6 % of firm-years in the dataset do not report any major customers. Among those that do report, the average number of customers is 3.5. Furthermore, 97.9 % of supplier-customer-year observations contribute more than 10 % to sales, while 2.1 % contribute less. In light of the ongoing discussion regarding the quality of different supply chain relationship datasets (Culot et al., 2023), we acknowledge the limitations associated with this data source, such as the reporting threshold that includes only buyers representing at least 10 % of the supplier's sales. This threshold may introduce sample selection bias, potentially excluding smaller but significant buyer-supplier relationships.

[6] To address potential biases introduced by the 10 % sales threshold, we conduct a robustness check by excluding all customers accounting for less than 10 % of sales, which allows us to just focus on the impact of significant supplier-customer relationships. The results remain unchanged (untabulated), confirming that minor customers do not skew our primary results. However, we still must acknowledge the potential limitations of our data, as this test only excludes disclosed minor customers; many undisclosed smaller customers, which are not captured by our dataset, remain untested.

[7] We also refer to the WRDS linktables available at https://wrds-www.wharton.upenn.edu/pages/wrds-research/database-linking-matrix/.

Campello and Gao (2017), the mean value of *Spread* is 5.1400, with a S.D. of 0.7564. In this sample, there are around 19 % observations with vertical common ownership. For common ownership measurements, the mean values of *CO_Firms, CO_Firms_MV, COwn_#* and *Avg_Investors* are 0.1286, 1.0950, 0.1178, and 0.0553, respectively. Statistics of other controls are quite similar to Graham et al. (2008), Campello and Gao (2017), and Murfin (2012).

4.2. Baseline regressions

Table 2 reports the regression results on the association between vertical common ownership and cost of loans based on eq. (1). The dependent variable loan spread (*Spread)* is used to measure the cost of loans at the facility level. The t-statistics reported in parentheses are calculated using both firm and year clustered heteroscedasticity robust standard error. Column (1) uses *CO_Firms* as a proxy for vertical common ownership. The coefficient is negative and significant at 5 % level (coefficient = −0.077, $t = -1.995$). For economic significance, a one S.D. increase in *CO_Firms* leads to a 0.47 percentage decrease in *Spread* (0.077 × 0.3135÷5.1400). The remaining columns are all significantly negative at 10 % level. As for control variables, firms with higher *Lev*, and lower *Size, MTB, Tangibility, Z_Score, LoanSize,* and non-performance pricing are associated with greater loan spreads, which is consistent with Graham et al. (2008) and Campello and Gao (2017). In summary, baseline regression results document vertical common ownership decreases loan spread.

4.3. The influence of hold-up problems

We then investigate the elements that may mitigate the negative effect of vertical common ownership on suppliers' cost of loans. First, considering the coordination channel, suppliers with significant relationship-specific investments are more vulnerable to hold-ups, especially if their customers switch to other suppliers, rendering these investments less valuable elsewhere. Previous research indicates that such firms are more likely to resort to accounting strategies to sustain their supply chain relationships (Raman and Shahrur, 2008). We argue that vertical common ownership could be particularly effective in reducing perceived supply chain risks among creditors, especially in cases where hold-up issues are pronounced. Therefore, the baseline association is anticipated to be stronger for businesses with higher relationship-specific investment. We use asset redeployability (*Redeploy*) measured by the BEA capital flow table (Kim and Kung, 2017) to proxy for relationship-specific investment. The subsample regression results based on asset redeployability are shown in Table 3. Firms are assigned to the low (high) group if the values of *Redeploy* are lower (higher) than the median.

In Table 3, the baseline results remain significantly negative when *Redeploy* = Low, but become insignificant when *Redeploy* = High. According to the *P*-values reported at the bottom of the panel, the differences between the two subsamples are significant at 1 % or 5 %. Overall, this cross-sectional test suggests that the baseline relationship is more significant for firms with greater relationship-specific investment.

4.4. The influence of bargaining power

Moreover, in line with the coordination role, we consider the baseline relationship to be more significant for suppliers with weaker bargaining power. Transactional agreements and their terms are largely influenced by the bargaining power of each party involved. Typically, the party with greater bargaining power secures more favorable contract terms (Hui et al., 2012). Suppliers with weaker bargaining power are more vulnerable to opportunistic actions from their customers. Therefore, vertical common ownership's role in coordination is likely more critical in these situations, potentially strengthening the baseline relationship. In this test, we use two variables to proxy for the bargaining power of the supplier firm and report the results in Table 4. The first one, industry entry rate (*Entry*), is calculated as the proportion of new firms to total firms within an industry, where the new firm is just established for one year or two years (Klapper et al., 2006). Higher industry entry rate means more suppliers are available for customers to choose from, where the bargaining power of suppliers is lower. Therefore, we argue that the baseline relationship is more pronounced for subsamples of *Entry* = High. This measurement is measured at the industry level. Besides, we also use a firm-level trade measurement to proxy for bargaining power. The trade credit provision (*AR*) is the accounts receivable of the supplier divided by lagged total assets. Suppliers with greater bargaining power tend to offer more credit financing to their customers (Fabbri and Klapper, 2016). Therefore, the baseline relationship is expected to be more pronounced when *AR* = High.

The subsample results are shown in Table 4, Panel A and Panel B. Firms are assigned to the low (high) group if the values of conditional variables are lower (higher) than the median. The negative baseline relationships remain significant when *Entry* = High and *AR* = High, but insignificant in the low groups. All of the within-group *P*-values are lower than 0.001 in Panel A, and lower than 0.10 in Panel B, which suggests that the negative baseline relationship is more pronounced for supplier firms with weaker bargaining power.

4.5. The influence of long-term common investors

Furthermore, we test whether the baseline relationship varies across common institutional investor types. Based on the monitoring hypothesis, the baseline relationship should be stronger for common institutional investors with more aptitude for monitoring. Research by Borochin and Yang (2017) suggests that long-term investors, unlike short-term investors focused on immediate gains, are more inclined to acquire in-depth information and engage in monitoring activities. Consequently, supplier firms with long-term institutional investors sharing ownership with their customers are likely under more effective monitoring, leading creditors to view

Table 1
Summary Statistics.

Variables	Mean	S. D.	P25	P50	P75
Spread	5.1400	0.7564	4.7958	5.3033	5.6384
CO_Firms	0.1286	0.3135	0.0000	0.0000	0.0000
CO_Firms_MV	1.0950	2.5942	0.0000	0.0000	0.0000
COwn_#	0.1178	0.2800	0.0000	0.0000	0.0000
Avg_Investors	0.0553	0.1378	0.0000	0.0000	0.0000
SIZE	6.6336	1.9705	5.1919	6.7055	8.0858
LEV	0.2932	0.1817	0.1519	0.2918	0.4246
MTB	1.6949	0.9710	1.1099	1.4088	1.9142
ROA	0.0333	0.1284	0.0000	0.0450	0.0886
Tangibility	0.3095	0.2493	0.1125	0.2310	0.4389
CFO_VOL	0.0680	0.0766	0.0218	0.0446	0.0844
Z_Score	1.9165	1.8031	0.8498	1.8182	2.8399
Rating	0.4718	0.4992	0.0000	0.0000	1.0000
Maturity	3.6893	0.6414	3.4012	3.9120	4.1109
LoanSize	4.5175	1.7475	3.2581	4.6151	5.8608
Performance	0.4049	0.4909	0.0000	0.0000	1.0000

Note: This table presents the descriptive statistics for variables used in the baseline analysis, the detailed variable definitions are presented in Appendix A.

Table 2
Vertical Common Ownership and Supplier's Cost of Loans.

	(1)	(2)	(3)	(4)
Dependent	*Spread*	*Spread*	*Spread*	*Spread*
CO_Firms	−0.077**			
	(−1.995)			
CO_Firms_MV		−0.009*		
		(−1.945)		
COwn_#			−0.076*	
			(−1.787)	
Avg_Investors				−0.157*
				(−1.848)
SIZE	−0.148***	−0.148***	−0.148***	−0.148***
	(−6.671)	(−6.661)	(−6.684)	(−6.678)
LEV	0.560***	0.556***	0.558***	0.559***
	(6.176)	(6.127)	(6.156)	(6.168)
MTB	−0.038**	−0.039**	−0.038**	−0.038**
	(−2.284)	(−2.315)	(−2.279)	(−2.253)
ROA	0.118	0.117	0.117	0.120
	(0.954)	(0.948)	(0.949)	(0.969)
Tangibility	−0.325**	−0.320**	−0.319**	−0.324**
	(−2.487)	(−2.457)	(−2.443)	(−2.483)
CFO_VOL	0.067	0.065	0.066	0.064
	(0.423)	(0.411)	(0.413)	(0.405)
Z_Score	−0.030**	−0.030**	−0.030**	−0.031**
	(−2.379)	(−2.370)	(−2.376)	(−2.401)
Rating	0.050	0.049	0.050	0.050
	(1.340)	(1.323)	(1.331)	(1.331)
Maturity	−0.028	−0.028	−0.029	−0.029
	(−1.525)	(−1.512)	(−1.532)	(−1.540)
LoanSize	−0.132***	−0.132***	−0.132***	−0.132***
	(−10.296)	(−10.305)	(−10.305)	(−10.309)
Performance	−0.097***	−0.097***	−0.097***	−0.097***
	(−4.409)	(−4.421)	(−4.412)	(−4.419)
Constant	5.413***	5.411***	5.414***	5.418***
	(13.116)	(13.104)	(13.122)	(13.127)
LoanPurpose FE		Yes		
LoanType FE		Yes		
Year FE		Yes		
Firm FE		Yes		
N	7005	7005	7005	7005
Adjusted R2	0.733	0.733	0.733	0.733

Note: This table reports the baseline regressions results. Variable definitions are presented in Appendix A. Significance at the 1 %, 5 %, and 10 % levels is indicated by ***, **, and *, respectively. The same below.

Table 3
Vertical Common Ownership, Hold-up Problems, and Cost of Loans.

Redeploy=	Low	High	Low	High	Low	High	Low	High
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Dependent	*Spread*		*Spread*		*Spread*		*Spread*	
CO_Firms	−0.112**	−0.041						
	(−1.978)	(−0.918)						
CO_Firms_MV			−0.012*	−0.005				
			(−1.948)	(−0.876)				
COwn_#					−0.128**	−0.024		
					(−2.025)	(−0.479)		
Avg_Investors							−0.233**	−0.075
							(−2.087)	(−0.743)
Constant	5.922***	6.477***	5.917***	6.477***	5.923***	6.475***	5.935***	6.478***
	(23.618)	(27.301)	(23.579)	(27.310)	(23.615)	(27.299)	(23.702)	(27.270)
Controls	Yes							
LoanPurpose FE	Yes							
LoanType FE	Yes							
Year FE	Yes							
Firm FE	Yes							
N	3359	3365	3359	3365	3359	3365	3359	3365
Adjusted R2	0.735	0.759	0.735	0.759	0.736	0.759	0.736	0.759
P	0.088		0.094		0.024		0.043	

Note: This table presents the results of regressing cost of loans on the vertical common ownership measurements partitioned by asset redeployability.

them as lower-risk entities.

We use the common ownership duration (*Duration*) counted in months of common investors as the proxy for long-term investors. Investors with higher duration own sufficient time or resources to monitor corporate behaviors more effectively (Chen et al., 2007). For firms with multiple common owners, we measure *Duration* by calculating the median value of investment lengths for all the common owners.[8] The subsample regressions are reported in Table 5. In this test, we just keep the sample with vertical common ownership. Within this sample, the average value of *Duration* is 54 months. The negative relationships between vertical common ownership and cost of loans remain significant when *Duration* = High, but become insignificant when *Duration* = Low. All of the *P*-values within groups are lower than 1 %, indicating that the baseline relationship between vertical common ownership and loan spread is more pronounced for commonly owned firms with long-term common institutional owners.

4.6. The influence of the information environment

In addition, to test the monitoring hypothesis, we investigate the effect of information asymmetry. According to Jin and Myers (2006) and Jensen and Meckling (2019), agency conflicts between borrowing firms and creditors become more severe when the company is operating in an opaque environment. Since we contend that vertical common ownership can effectively monitor supply chain partners' misconduct and lower agency costs, thereby reducing the cost of loans, we hypothesize that the role of vertical common ownership may be more prominent for firms with higher information asymmetry.

We use the information opacity measure to proxy for the information environment. *Opacity* is calculated as the prior three years' moving sum of the absolute value of discretionary accruals (Hutton et al., 2009). The subsample regressions are reported in Table 6. The negative baseline relationships remain significant when *Opacity* = High, but become insignificant when *Opacity* = Low. All of the *P*-values within groups are lower than 5 %, indicating that the baseline relationship is more pronounced for firms with higher information asymmetry.

4.7. The influence of financial position

We further test the mediation effect of corporate financial conditions. A firm's financial condition is a vital determinant of its solvency or default risk. External supervision becomes more important for a firm with poorer financial conditions (Fields et al., 2012). Thus, creditors will commend vertical common ownership more in terms of lower cost of loans. Moreover, we argue that supplier firms with weaker financial positions are more vulnerable to customers' hold-ups, and thus have stronger incentives to engage in accounting manipulation, where the coordination role of common institutional investors may be more important. According to the argument above, the influence of financial condition provides evidence for the overall effect of vertical common ownership on the cost of loans, so we do not distinguish between coordination and monitoring channels in this test. To sum up, we expect the relation of common ownership with reduced cost of loans to be stronger for firms with weaker financial situations.

[8] We have not use Bushee's institutional classification in this study, because the sample size is quite limited for facility data. There are still many missing data in Bushee classification, therefore the data will be even less.

Table 4
Vertical Common Ownership, Bargaining Power, and Cost of loans.

Panel A: Entry Rate								
Entry=	Low	High	Low	High	Low	High	Low	High
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Dependent	*Spread*		*Spread*		*Spread*		*Spread*	
CO_Firms	0.019	−0.246***						
	(0.406)	(−3.770)						
CO_Firms_MV			0.004	−0.031***				
			(0.730)	(−4.196)				
COwn_#					0.040	−0.294***		
					(0.810)	(−4.314)		
Avg_Investors							0.089	−0.505***
							(0.927)	(−3.767)
Constant	5.686***	5.640***	5.693***	5.669***	5.686***	5.655***	5.678***	5.656***
	(17.389)	(12.767)	(17.488)	(12.790)	(17.430)	(12.800)	(17.367)	(12.751)
Controls				Yes				
LoanPurpose FE				Yes				
LoanType FE				Yes				
Year FE				Yes				
Firm FE				Yes				
N	3795	3210	3795	3210	3795	3210	3795	3210
Adjusted R2	0.756	0.775	0.756	0.775	0.756	0.775	0.756	0.775
P	<0.001		<0.001		<0.001		<0.001	
Panel B: Trade Credit								
AR=	Low	High	Low	High	Low	High	Low	High
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Dependent	*Spread*		*Spread*		*Spread*		*Spread*	
CO_Firms	−0.049	−0.156***						
	(−0.944)	(−2.718)						
CO_Firms_MV			−0.007	−0.018***				
			(−1.145)	(−2.590)				
COwn_#					−0.035	−0.176***		
					(−0.607)	(−2.750)		
Avg_Investors							−0.153	−0.316**
							(−1.480)	(−2.261)
Constant	5.718***	6.206***	5.715***	6.195***	5.720***	6.205***	5.716***	6.202***
	(16.213)	(27.016)	(16.180)	(27.052)	(16.239)	(27.066)	(16.136)	(27.015)
Controls				Yes				
LoanPurpose FE				Yes				
LoanType FE				Yes				
Year FE				Yes				
Firm FE				Yes				
N	3588	3417	3588	3417	3588	3417	3588	3417
Adjusted R2	0.728	0.754	0.728	0.754	0.728	0.754	0.728	0.754
P	0.063		0.096		0.039		0.087	

Note: This table presents the results partitioned by the median of the supplier firm's bargaining power. Panel A reports the results partitioning the sample based on industry entry rate (*Entry*). The partition variable of Panel B is the trade credit (*AR*).

We use Altman (1968) *Z*-Score to proxy for corporate financial position. The lower the value of *Z_Score*, the weaker of firm's financial position. The subsample regression results based on financial position are shown in Table 7. As can be seen, the baseline relationship is stronger for firms with lower *Z_Score*, and all of the P-values within groups are lower than 0.10, suggesting that the baseline relation is stronger for firms with poorer financial positions.

4.8. Additional tests

Lastly, we proceed to unveil whether other financial variables may also change in response to vertical common ownership.[9] Notably, vertical common ownership fosters more efficient supply chain relationships, as indicated by enhanced trade credit provision and lowered inventory levels. In addition, since vertical common ownership lowers creditors' perception of supply chain risk,

[9] Due to constraints on word count, detailed results are not included.

Table 5
Vertical Common Ownership, Long-term Common Investors, and Cost of loans.

Duration=	0	1	0	1	0	1	0	1
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Dependent	*Spread*		*Spread*		*Spread*		*Spread*	
CO_Firms	0.287	−0.312***						
	(0.893)	(−3.054)						
CO_Firms_MV			0.050	−0.044***				
			(0.563)	(−3.362)				
COwn_#					0.862	−0.343***		
					(0.844)	(−2.756)		
Avg_Investors							0.236	−0.395*
							(0.423)	(−1.880)
Constant	5.296***	3.875***	5.114***	3.780***	5.298***	3.860***	5.394***	3.839***
	(3.980)	(4.108)	(4.026)	(3.925)	(3.991)	(4.038)	(4.222)	(3.945)
Controls				Yes				
LoanPurpose FE				Yes				
LoanType FE				Yes				
Year FE				Yes				
Firm FE				Yes				
N	629	665	629	665	629	665	629	665
Adjusted R2	0.801	0.756	0.802	0.757	0.801	0.755	0.803	0.750
P	0.002		<0.001		<0.001		0.001	

Note: This table presents the results partitioned by common institutional investors' types. The long-term investor is measured by the duration of ownership (*Duration*) calculated in months.

Table 6
Vertical Common Ownership, Information Environment, and Cost of Loans.

Opacity =	Low	High	Low	High	Low	High	Low	High
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Dependent	*Spread*		*Spread*		*Spread*		*Spread*	
CO_Firms	−0.010	−0.157***						
	(−0.166)	(−2.680)						
CO_Firms_MV			−0.003	−0.019***				
			(−0.348)	(−2.701)				
COwn_#					−0.021	−0.166**		
					(−0.307)	(−2.389)		
Avg_Investors							0.111	−0.378***
							(0.820)	(−2.955)
Constant	5.006***	5.531***	5.005***	5.512***	5.007***	5.527***	4.999***	5.568***
	(9.655)	(22.247)	(9.658)	(21.868)	(9.659)	(22.125)	(9.625)	(22.481)
Controls				Yes				
LoanPurpose FE				Yes				
LoanType FE				Yes				
Year FE				Yes				
Firm FE				Yes				
N	3411	3594	3411	3594	3411	3594	3411	3594
Adjusted R2	0.790	0.750	0.790	0.750	0.790	0.750	0.790	0.750
P	0.011		0.026		0.034		<0.001	

Note: This table presents the results partitioned by the median of the supplier firm's information environment, as measured by information opacity.

subsequent corporate financing and investment levels will increase intuitively. Empirical results support these arguments.[10]

5. Robustness

5.1. Endogeneity issues

While the initial regression indicates a negative correlation between common ownership and the cost of loans for supplier firms, there are potential issues of causality in these findings. This could be due to variables not accounted for or reverse causality. For

[10] Specifically, we measure corporate financing as net debt and equity issuance, then divided by lagged total assets (Shroff, 2017). We choose capital expenditures and research and development expenses as investment proxies.

Table 7
Vertical Common Ownership, Financial Position, and Cost of Loans.

Z_Score =	Low	High	Low	High	Low	High	Low	High
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Dependent	*Spread*		*Spread*		*Spread*		*Spread*	
CO_Firms	−0.129**	−0.030						
	(−2.495)	(−0.597)						
CO_Firms_MV			−0.016***	−0.004				
			(−2.683)	(−0.596)				
COwn_#					−0.145**	−0.029		
					(−2.480)	(−0.508)		
Avg_Investors							−0.258**	−0.051
							(−2.347)	(−0.441)
Constant	7.281***	6.352***	7.264***	6.353***	7.288***	6.355***	7.240***	6.357***
	(24.569)	(19.850)	(24.478)	(19.842)	(24.495)	(19.843)	(24.473)	(19.844)
Controls				Yes				
LoanPurpose FE				Yes				
LoanType FE				Yes				
Year FE				Yes				
Firm FE				Yes				
N	3387	3307	3387	3307	3387	3307	3387	3307
Adjusted R2	0.776	0.717	0.776	0.717	0.776	0.717	0.776	0.717
P	0.076		0.046		0.031		0.038	

Note: This table presents the results partitioned by the supplier's financial position, measured by Altman *Z-Score*.

Table 8
Endogeneity Analyses.

Panel A: Difference-in-Difference Analysis

	(1)	(2)	(3)	(4)
Dependent	*Spread*	*Spread*	*Spread*	*Spread*
*Treat*Post*	−0.244***	−0.198*	−0.194**	−0.173*
	(−2.85)	(−1.91)	(−2.05)	(−1.76)
Constant	4.433***	5.643***	5.044***	5.688***
	(29.61)	(5.97)	(36.31)	(20.70)
Control	No	Yes	No	Yes
LoanPurpose	Yes	Yes	Yes	Yes
LoanType	Yes	Yes	Yes	Yes
Year×Merger FE	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	No
N	960	879	536	518
Adjusted R2	0.835	0.859	0.828	0.866

Panel B: Excluding Banks Involved in M&As

	(1)	(2)	(3)	(4)
Dependent	*Spread*	*Spread*	*Spread*	*Spread*
CO_Firms	−0.092*			
	(−1.881)			
CO_Firms_MV		−0.012**		
		(−2.196)		
COwn_#			−0.088*	
			(−1.649)	
Avg_Investors				−0.184*
				(−1.939)
Constant	5.642***	5.638***	5.645***	5.646***
	(13.618)	(13.603)	(13.630)	(13.628)
Controls	Yes	Yes	Yes	Yes
LoanPurpose FE	Yes	Yes	Yes	Yes
LoanType FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes
N	5393	5393	5393	5393
Adjusted R2	0.744	0.745	0.744	0.744

Note: Panel A presents the DiD analysis based on M&As between financial institutions. Columns (3) and (4) in Panel A further exclude M&As involving banks. We discard firm fixed effects in column (4) Panel A because many firm-level controls are omitted due to the small sample size. Panel B reports baseline regressions after excluding all banks involved in M&A activities.

instance, institutional investors might prefer firms with a lower risk of default, and the observed relationship may also stem from unobserved common factors. To determine the causal impact of common ownership, we conduct a DiD analysis, utilizing M&As among financial institutions as exogenous shocks to common ownership, to more accurately assess this relationship.

Following He and Huang (2017), we identify the total financial institution M&As between financial institutions (i.e., both the acquirers and targets are financial institutions). We then match both the acquirers and targets of all such M&As to the 13F database to identify merging institutions reported in the 13F. This procedure yields a list of M&As where both the acquirers and targets are 13F institutional investors. We construct the sample of treated firms as follows. For each firm held by an acquirer (target) in the quarter just before the merger announcement date, the firm is identified as a treated firm if the merging target (acquirer) blockholds any of the major customers just before the announcement date. In other words, a treated firm is defined as a firm that experiences an increase in vertical common ownership due to the mergers of institutional investors. For each treated firm, a non-treated firm is selected as the matched control firm if i) it was blockheld by the same merging institution before the merger, and ii) it did not have major customers blockheld by the other merging institution. Finally, there is a total of sixteen financial institutions' M&As in the sample. We then perform the DiD estimation in the following Eq. (2).

$$\begin{aligned} Spread_{i,j,k,t} = & \alpha + \beta \times Treat_{i,j} \times Post_{j,t} + \gamma' FirmControls_{i,j,t} + \delta' LoanControls_{k,j,t} \\ & + Loan\ Purpose\ FE + Loan\ Type\ FE + Year \times Merger\ FE + Firm\ FE + \varepsilon_{i,j,k,t} \end{aligned} \tag{2}$$

Here, *i, j, k, t* represent firm *i*, financial institution merger *j*, facility *k*, and year *t*, respectively. *Treat* is a dummy variable that equals one if a firm *i* is a treated firm in merger *j* (i.e., it experienced an increase in common ownership with its major customers due to the

Table 9
Vertical Common Ownership versus Horizontal Common Ownership.

Panel A: Intergroup Difference Analysis

		(1)	(2)	(2)–(1)
	Spread	*VCO* = 0	*VCO* = 1	*Diff*
(3)	*HCO* = 0	5.1383	4.9484	−0.1899**
(4)	*HCO* = 1	5.1705	5.0765	−0.0940**
(4)–(3)	*Diff*	0.0322	0.1281	

Panel B: Regression Tests

	(1)	(2)	(3)	(4)
Dependent	*Spread*	*Spread*	*Spread*	*Spread*
CO_Firms	−0.084**			
	(−1.977)			
HCO_Firms	0.006			
	(0.405)			
CO_Firms_MV		−0.009*		
		(−1.773)		
HCO_Firms_MV		−0.001		
		(−0.303)		
COwn_#			−0.086*	
			(−1.878)	
HCOwn_#			0.016	
			(0.580)	
Avg_Investors				−0.098**
				(−1.979)
HAvg_Investors				0.004
				(0.216)
INST	0.011	0.011	0.015	0.008
	(0.197)	(0.265)	(0.137)	(0.227)
Constant	5.409***	5.411***	5.410***	5.411***
	(13.081)	(13.113)	(13.083)	(13.094)
Controls		Yes		
LoanPurpose FE		Yes		
LoanType FE		Yes		
Year FE		Yes		
Firm FE		Yes		
N	7005	7005	7005	7005
Adjusted R2	0.733	0.733	0.733	0.733

Note: This table compares vertical common ownership and horizontal common ownership. Panel A lists the average value of *Spread* in four groups (*VCO* = 0 and *HCO* = 0, *VCO* = 1 and *HCO* = 0, *VCO* = 0 and *HCO* = 1, *VCO* = 1 and *HCO* = 1) and the intergroup differences among every two groups. Panel B shows the regression tests adding vertical common ownership, horizontal common ownership and institutional investor shareholdings altogether.

M&As of financial institutions), and zero otherwise. We use a three-year window around each financial institution's M&A as the sample. *Post* is a dummy that takes a value of one in the three years after the M&As of financial institutions, and zero in the three years before. We include controls similar to those used in the baseline panel regression. There may exist stacked data, because a facility may be treated group in a financial institution M&A, but work as a control group in another. Therefore, we control for year×merger fixed effect and firm fixed effect in the regressions.

Columns (1) and (2) of Table 8 Panel A show the DiD results on the effect of vertical common ownership on suppliers' cost of loans based on M&As between financial institutions. We regress *Spread* on *Treat*×*Post*, and all the fixed effects in column (1), and also add other controls in column (2). As can be seen, coefficients of *Treat*×*Post* are negatively significant, supporting baseline relationships.

Furthermore, there may be a concern that many financial institution mergers involve banks. Consequently, these mergers also represent a shock to the loan market, suggesting that changes in loan spreads might not solely result from shifts in common ownership but also from direct impacts on lenders' behaviors. To address this complexity and distinguish the effects of common ownership from those of lender-specific shocks, we have implemented two more checks. First, we refine our DiD approach by excluding M&As involving banks to isolate the effect of common ownership from the potential confounding impacts of financial institution mergers. This adjustment reduces our sample size from 960 to 536, cutting it nearly in half. The results are shown in the column (3) and (4) of Table 8 Panel A, which remain unchanged. Then, we re-estimate our main regression analysis by excluding all banks involved in M&A activities, ensuring that our results are not influenced by direct changes in loan pricing strategies due to these mergers. The results are reported in Table 8 Panel B, which are still consistent and affirm our findings. Overall, these two additional tests further alleviate endogeneity concerns, demonstrating that our findings are robust against variations in the sample caused by bank mergers.

5.2. Vertical common ownership versus horizontal common ownership

Further, there may be a concern that vertical common ownership acts in the same role as horizontal one, which means the firm has common institutional investors with its competing peers. To differentiate the effect of vertical common ownership from horizontal one, we conduct two robustness tests in this section.

First, referring to Park et al. (2019), we construct dummy variables *VCO* (and *HCO*), which equal one if a firm has common ownership with its customers (industry peers) for more than three quarters within the fiscal year, otherwise zero. In this section, the industry competing peers are identified by the 4-digit SIC industry classification code. Results remain similar if we use a 2 or 3-digit SIC code. As in the baseline regression, we also set 5 % as the shares' cutoff for common ownership measurements construction. Statistics show that the mean value of *VCO* is 0.12, and the mean value of *HCO* is 0.45, which is more prevalent. Then, based on the values of *VCO* and *HCO*, we conduct a 2×2 matrix to compare the intergroup differences of loan spread *(Spread)*. The results are shown in Panel A of Table 9. Panel A lists the average value of *Spread* in the four groups ($VCO = 0$ and $HCO = 0$, $VCO = 1$ and $HCO = 0$, $VCO = 0$ and $HCO = 1$, $VCO = 1$ and $HCO = 1$) and the intergroup differences among every two groups. For any value of *HCO*, the differences between $VCO = 1$ and $VCO = 0$ are significantly negative at 5 % level. However, the differences between $HCO = 1$ and $HCO = 0$ are always insignificantly positive. To sum up, horizontal common ownership has no effect in lowering the cost of loans in supply chain settings, while the role of vertical common ownership is still significant, even ruling out the influence of horizontal one.

Second, we set a series of horizontal common ownership measures similar to the vertical ones in baseline tests and add them together in the regressions. Specifically, *HCO_Firms* is the natural log of one plus the number of the firm's industry peers that are co-owned with the firm. *HCO_Firms_MV* is the natural log of one plus the market value of the firm's industry peers that are co-owned with the firm. *HCOwn_#* is the natural log of one plus the number of institutional investors who are blockholders of both the firm and its industry peers. *HAvg_Investors* is the number of horizontal common institutional investors scaled by the number of total institutional investors in the firm. We also add the percentage of shares held by institutional investors *(INST)* in the regression. Results of these two kinds of common ownership measurements and institutional investor shareholdings are reported in Panel B of Table 9. It shows that the coefficients of vertical common ownership measures are still significant at 5 % or 1 % level, while the horizontal ones are far from significant; some are even positive. For the only negative coefficient of horizontal common ownership measurement in column (2), the coefficient magnitude of vertical common ownership (−0.009) is 9 times larger than that of horizontal common ownership (−0.001). Vertical common ownership definitely predominates. In addition, coefficients of *INST* are neither significant. Overall, it illustrates the incremental effect of vertical common ownership compared to horizontal common ownership and institutional ownership.

5.3. Supplier-customer pair level analysis

This paper studies the role of vertical common ownership on the supplier side in baseline regressions. We test our research question based on the supplier-customer dyad level in this section for robustness check.

We set three variables of interest to measure vertical common ownership for a supplier-customer dyad. The first one is *COwn_Pair*, a dummy variable which equals one if a supplier-customer dyad has any common ownership, otherwise zero. The second one is *Sum_COwn_Pair*, calculated as the natural log value of one plus the number of common institutional owners for a supplier-customer dyad. The third one, *COwn_Value_Pair*, is the percentage of market value held by common institutional investors for a supplier-customer dyad, which equals the total market value of shares for which the supplier-customer dyad is commonly held, then divided by the total market value of the supplier firm and customer firm. Since institutional shareholdings are provided by quarter data in the Thomson database, the above vertical common ownership measurements are calculated in their average value within a fiscal year. In this test, we control for loan purpose, loan type, year, supplier firm and customer firm fixed effects in all regressions. Results are reported in Table 10, where the observations increase to 9592. The coefficients of vertical common ownership pairs are all

significantly negative at 5 %, which further strengthens the findings.

5.4. Alternative vertical common ownership measurements

We further conduct robustness checks using several other vertical common ownership measurements. First, since our analysis is conducted at the supplier level, it is crucial to understand how much of the supplier's supply chain is exposed to common ownership. Therefore, firm-based common ownership measurements, which reflect the number of a supplier's customers that share a major institutional investor with the supplier, should be more appropriate. Accordingly, we have adjusted our initial measurements in the baseline regression model to be scaled by the total number of customers each firm has. Specifically, *CO_AvgFirms* represents the number of the firm's customers that are co-owned with the supplier, scaled by the total number of customers that the firm has. Similarly, *CO_AvgFirms_MV* represents the market value of the firm's customers that are co-owned with the supplier, scaled by the total market value of all the firm's customers. The regression results, presented in columns (1) and (2) of Table 11, remain consistent with previous findings.

Additionally, following He and Huang (2017) and Freeman (2023), we consider a series of other vertical common ownership measurements. The first one is *COwn_D*, which equals one if a firm has any common ownership with its customers, otherwise zero. The second is *COwn_Avg_EW*, which is the arithmetic average of the supplier firm *i*'s shares and customer *j*'s shares held by common investors. It is aggregated at the firm level by using an equal-weighted average across all its customers. We also use the market value-weighted average value (*COwn_Avg_VW*) and sales-weighted average value (*COwn_Avg_SW*) for robustness tests. In addition, we set the variable *COwn_Value_EW*, which is the total market value held by all the common investors of the supplier firm *i* and its customer *j*, scaled by the total market capitalization of supplier *i* and customer *j*. It is aggregated at the supplier firm level by using an equal-weighted average across all of its customers. We also use the market value-weighted average value (*COwn_Value_VW*) and sales-weighted average value (*COwn_Value_SW*) for robustness tests. Since institutional shareholdings are provided by quarter data in the Thomson database, the above vertical common ownership measurements are averaged within a fiscal year. Regression results are reported in the remaining columns of Table 11. The coefficients of all the vertical common ownership measures are significantly negative at 1 % or 5 %, which further supports our arguments.

5.5. Alternative cost of loans measurements

In this section, we use other measures of cost of loans (Melnik and Plaut, 1986). The first is loan maturity (*Maturity*), is the natural log value of one plus loan facility maturity counted in months. The second one is loan amount (*LoanSize*), equals the natural log value of one plus the total loan amount, measured in millions of dollars.[11] The third one is the number of lenders (*No_Lenders*), which equals the natural log value of one plus the number of lenders in a loan facility. The regression results are reported in Panel A (*Maturity*), Panel B (*LoanSize*), and Panel C (*NoLenders*) of Table 12, respectively. Most of the coefficients of vertical common ownership measurements are significantly positive for loan maturity, loan amount, and the number of lenders (only slightly insignificant for *Avg_Investors* in Panel B and Panel C), providing further evidence that besides loan spread, vertical common ownership can also decrease the other dimensions of cost of loans.

Lastly, there may be concerns that relying solely on all-in spread drawn from drawn portion of loan facilities might be incomplete, as the undrawn portion of a credit line also incurs spread/fees. Following the methodology outlined by Berg et al. (2016), we have expanded our analysis to include the total cost of borrowing (*TCB*) as a measure of loan costs. *TCB* equals (*Upfront Fee / Expected Loan Maturity in Years*) + (1-*PDD*) * (*Facility Fee* + *Commitment Fee*) + *PDD* * (*Facility Fee* + *Spread*) + *PDD* * *Prob(Utilization* > *UtilizationThreshold* | *Usage* > *0)* * *Utilization Fee* + *Prob(Cancellation)* * *Cancellation Fee*. In the equation, the *Upfront Fee* represents a one-time payment made by the borrower at the loan's closing date. The *Expected Loan Maturity in Years* is the duration between the facility's start and end dates. *PDD* stands for the probability of a drawdown, indicating the likelihood that the borrower will draw on the credit line. The *Facility Fee* is an annual fee charged on the total committed amount, regardless of usage, while the *Commitment Fee* is paid on the unused portion of the loan commitment. The *Spread* is the interest margin above the interbank loan rate for the drawn portion of the loan. *Prob(Utilization* > *UtilizationThreshold* | *Usage* > *0)* is the probability that the utilization rate will exceed a specified threshold, given that there is usage, and the *Utilization Fee* is charged when this threshold is exceeded. Finally, *Prob(Cancellation)* represents the probability that the loan will be canceled before maturity, and the *Cancellation Fee* is the cost associated with such an early termination.[12] The regression results are presented in Table 12 Panel D, which are in line with our findings.

[11] We thank the anonymous referee for suggesting that we explore the extensive margin related to loan approvals and vertical common ownership. Unfortunately, our dataset does not include data on the approval status of loan applications. Instead, we analyze the total amount of loans received by firms, which allows us to indirectly assess the impact of vertical common ownership on financial access. This result aligns with our baseline findings and underscores the positive role of vertical common ownership in enhancing financial access at the extensive margin. We acknowledge the limitation due to data unavailability.

[12] We thank Professor Tobias Berg for making this data and code publicly available. See https://sites.google.com/view/tobias-berg/startseite/data-and-code. However, we do not use *TCB* as the baseline dependent variable due to the higher incidence of missing values.

Table 10
Pair Level Analysis.

	(1)	(2)	(3)
Dependent	*Spread*	*Spread*	*Spread*
COwn_Pair	−0.088**		
	(−2.555)		
COwn_#_Pair		−0.096**	
		(−2.251)	
COwn_Value_Pair			−0.813**
			(−2.448)
Constant	5.459***	5.459***	5.455***
	(9.041)	(9.041)	(9.029)
Control		Yes	
LoanPurpose FE		Yes	
LoanType FE		Yes	
Year FE		Yes	
Supplier Firm FE		Yes	
Customer Firm FE		Yes	
N	9592	9592	9592
Adjusted R2	0.764	0.763	0.764

Note: This table reports the supplier-customer pair-level regression results.

Table 11
Alternative Common Ownership Measurements.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Dependent	*Spread*	*Spread*	*Spread*	*Spread*	*Spread*	*Spread*	*Spread*	*Spread*	*Spread*
CO_AvgFirms	−0.170**								
	(−2.440)								
CO_AvgFirms_MV		−21.694**							
		(−2.515)							
COwn_D			−0.064*						
			(−1.821)						
COwn_Avg_EW				−1.054*					
				(−1.940)					
COwn_Avg_VW					−1.016*				
					(−1.946)				
COwn_Avg_SW						−1.029*			
						(−1.888)			
COwn_Value_EW							−2.144*		
							(−1.924)		
COwn_Value_VW								−2.039*	
								(−1.896)	
COwn_Value_SW									−2.119*
									(−1.902)
Constant	60.292***	56.766***	7.532***	7.530***	7.534***	7.532***	7.533***	7.536***	7.534***
	(2.955)	(2.721)	(21.392)	(21.363)	(21.366)	(21.358)	(21.373)	(21.369)	(21.369)
Controls					Yes				
LoanPurpose FE					Yes				
LoanType FE					Yes				
Year FE					Yes				
Firm FE					Yes				
N	7005	7005	7005	7005	7005	7005	7005	7005	7005
Adjusted R2	0.719	0.718	0.733	0.733	0.733	0.733	0.733	0.733	0.733

Note: This table shows robustness tests on other independent variables.

5.6. The drivers of vertical common ownership and Heckman analysis

Given that vertical common ownership is an endogenously determined variable and scant literature exists exploring its determinants, concerns may arise regarding what drives variation in vertical common ownership. Our study aims to address this gap to some extent. We propose that the primary drivers of vertical common ownership could stem from two sources: the fraction of customer sales and the share ownership held by institutional investors. First, fraction of customer sales reflects a supplier's dependency on each customer. A higher proportion of sales to a particular customer not only highlights strategic importance of that relationship but also encourages the alignment of objectives through common ownership. Such alignment is expected to smooth transactional flows, enhance trust, and foster stable long-term relationships (Shi, 2016), thereby increasing the likelihood of vertical common ownership as a strategic choice to secure and enhance these critical business engagements. Second, we suggest that companies with greater

Table 12
Alternative Cost of loans Measurements.

Panel A: Loan Maturity

	(1)	(2)		(3)	(4)
Dependent	*Maturity*	*Maturity*		*Maturity*	*Maturity*
CO_Firms	0.091***				
	(2.926)				
CO_Firms_MV		0.012***			
		(3.317)			
COwn_#				0.104***	
				(3.034)	
Avg_Investors					0.159**
					(2.387)
Constant	2.774***	2.775***		2.771***	2.770***
	(11.902)	(11.914)		(11.905)	(11.885)
Controls			Yes		
LoanPurpose FE			Yes		
LoanType FE			Yes		
Year FE			Yes		
Firm FE			Yes		
N	7005	7005		7005	7005
Adjusted R2	0.584	0.584		0.584	0.583

Panel B: Loan Amount

	(1)	(2)		(3)	(4)
Dependent	*LoanSize*	*LoanSize*		*LoanSize*	*LoanSize*
CO_Firms	0.180**				
	(2.329)				
CO_Firms_MV		0.018**			
		(2.010)			
COwn_#				0.173*	
				(1.940)	
Avg_Investors					0.250
					(1.624)
Constant	−0.125	−0.121		−0.128	−0.133
	(−0.430)	(−0.415)		(−0.441)	(−0.456)
Controls			Yes		
LoanPurpose FE			Yes		
LoanType FE			Yes		
Year FE			Yes		
Firm FE			Yes		
N	7005	7005		7005	7005
Adjusted R2	0.869	0.868		0.868	0.868

Panel C: Number of Lenders

	(1)	(2)		(3)	(4)
Dependent	*No_Lenders*	*No_Lenders*		*No_Lenders*	*No_Lenders*
CO_Firms	0.105**				
	(2.250)				
CO_Firms_MV		0.011*			
		(1.888)			
COwn_#				0.107**	
				(2.032)	
Avg_Investors					0.155
					(1.559)
Constant	−0.529***	−0.526***		−0.531***	−0.534***
	(−2.837)	(−2.822)		(−2.849)	(−2.861)
Controls			Yes		
LoanPurpose FE			Yes		
LoanType FE			Yes		
Year FE			Yes		
Firm FE			Yes		
N	7005	7005		7005	7005
Adjusted R2	0.702	0.702		0.702	0.702

(*continued on next page*)

Table 12 (*continued*)

Panel C: Number of Lenders				
	(1)	(2)	(3)	(4)
Dependent	*No_Lenders*	*No_Lenders*	*No_Lenders*	*No_Lenders*

Panel D: Total Cost of Borrowing				
	(1)	(2)	(3)	(4)
Dependent	*TCB*	*TCB*	*TCB*	*TCB*
CO_Firms	−35.536***			
	(−3.360)			
CO_Firms_MV		−3.525***		
		(−3.908)		
COwn_#			−36.151***	
			(−3.577)	
Avg_Investors				−56.698***
				(−3.651)
Constant	232.479***	231.992***	231.898***	231.240***
	(6.392)	(6.379)	(6.376)	(6.385)
Controls		Yes		
LoanPurpose FE		Yes		
LoanType FE		Yes		
Year FE		Yes		
Firm FE		Yes		
N	3167	3167	3167	3167
Adjusted R2	0.748	0.748	0.748	0.748

Note: This table shows the results on alternative loan characteristics.

institutional investor share ownership often possess more vertical common ownership. This trend is likely due to the inherent characteristics of these firms, such as robust governance structures, operational transparency, and strong financial performance, which attract institutional investors (Boone and White, 2015). Consequently, firms favored by institutional investors for these traits tend to exhibit higher levels of vertical common ownership.

Furthermore, we utilize Heckman model to account for these determinants. Given the need to control for the fraction of customer sales, this analysis is conducted at the supplier-customer dyad level. We firstly estimate Probit selection equation to determine the likelihood of vertical common ownership based on these potential determinants. Specifically, we regress the vertical common ownership dummy (*COwn_Pair*) on the fraction of customer sales (*Sale_Ratio*), the share of institutional investors' ownership (*INST*), and other control variables. This yields the Inverse Mills Ratio (*IMR*). Moreover, coefficients of *Sale_Ratio* and *INST* are both significantly positive, consistent with our arguments above. The second stage involves correcting for any selection bias by incorporating the *IMR* as an additional control. Results are shown in Table 13. The measures of vertical common ownership remain significant, thereby corroborating our results.[13]

6. Conclusions

This paper presents evidence linking vertical common ownership with cost of loans for supplier firms. We find that creditors view suppliers with vertical common owners as less likely to default, as indicated by lower loan spreads. This negative correlation is more distinct in suppliers with greater relationship-specific investments, weaker bargaining power, higher information asymmetry, long-term common institutional owners, and weaker financial positions. The robustness of these results is confirmed through various methods, including analyses of financial institution M&As, supplier-customer pair level assessments, and incorporating horizontal common ownership and institutional ownership as control variables, along with other tests. Additionally, the study explores the impact of vertical common ownership on other financial metrics, revealing an increase in subsequent financing and investment activities, including capital and R&D expenditures. We also find that the fraction of customer sales and the institutional investors' share ownership are likely to influence the likelihood of vertical common ownership. Our results consistently hold under the Heckman analysis.

These findings align with the argument that institutional investors with vertical common ownership enhance firm coordination and monitoring, thereby reducing risks associated with hold-ups, opportunism, and adverse selection, ultimately leading to lower risk premiums from creditors. This research illuminates the role of common ownership in supply chains and its effect on reducing the cost of loans, offering valuable insights for corporate management, regulators, and external creditors. It underscores the significance of

[13] However, we acknowledge the limitation that our analysis does not capture all possible factors influencing vertical common ownership. The company characteristics, sales proportions, and ownership structures considered here represent only a partial view. Further research is necessary to explore additional determinants that could significantly impact vertical common ownership.

Table 13
Heckman Analysis.

	(1)	(2)	(3)	(4)
Dependent	*COwn_Pair*	*Spread*	*Spread*	*Spread*
Sale_Ratio	0.261*	−0.116*	−0.114*	−0.111*
	(1.759)	(−1.755)	(−1.718)	(−1.661)
INST	1.646***	−0.369	−0.370	−0.361
	(16.310)	(−1.213)	(−1.203)	(−1.179)
IMR		−0.125	−0.118	−0.110
		(−0.564)	(−0.528)	(−0.492)
COwn_Pair		−0.090***		
		(−2.665)		
COwn_#_Pair			−0.084**	
			(−2.001)	
COwn_Value_Pair				−0.737**
				(−2.298)
Constant	−1.197***	6.323***	6.298***	6.265***
	(−4.103)	(8.037)	(7.953)	(7.934)
Control	Yes	Yes	Yes	Yes
LoanPurpose FE	Yes	Yes	Yes	Yes
LoanType FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Supplier Firm FE	No	Yes	Yes	Yes
Customer Firm FE	No	Yes	Yes	Yes
N	6525	6525	6525	6525
Adjusted R2 (Pseudo R2)	0.294	0.776	0.776	0.776

Note: This table reports the Heckman two-stage analysis. We omit firm fixed effects in the first stage in column (1) because non-linear models like Probit can yield inconsistent point estimates due to the incidental parameter problem (Angrist and Pischke, 2009).

vertical common ownership in both operational coordination and financial stability.

CRediT authorship contribution statement

Haowen Tian: Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Junkai Wang:** Writing – review & editing, Visualization, Validation, Supervision, Software. **Sirui Wu:** Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Supervision, Validation, Writing – review & editing.

Data availability

Data will be made available on request.

Appendix A. Variable Definitions

Dependent Variables	
Spread	The natural log of the all-in spread drawn charged by the bank over LIBOR for the drawn portion of the loan facility.
Maturity	The natural log of loan facility maturity counted in months.
LoanSize	The natural log of loan facility total amount, measured in millions of dollars.
NoLenders	The natural log of one plus the number of lenders in a loan.
TCB	The total cost of borrowing, equals (*Upfront Fee / Expected Loan Maturity in Years*) + (1-*PDD*) * (*Facility Fee* + *Commitment Fee*) + *PDD* * (*Facility Fee* + *Spread*) + *PDD* * *Prob(Utilization* > *UtilizationThreshold* | *Usage* > *0)* * *Utilization Fee* + *Prob(Cancellation)* * *Cancellation Fee*. In the equation, the *Upfront Fee* represents a one-time payment made by the borrower at the loan's closing date. The *Expected Loan Maturity in Years* is the duration between the facility's start and end dates. *PDD* stands for the probability of a drawdown, indicating the likelihood that the borrower will draw on the credit line. The *Facility Fee* is an annual fee charged on the total committed amount, regardless of usage, while the *Commitment Fee* is paid on the unused portion of the loan commitment. The *Spread* is the interest margin above the interbank loan rate for the drawn portion of the loan. *Prob(Utilization* > *UtilizationThreshold* | *Usage* > *0)* is the probability that the utilization rate will exceed a specified threshold, given that there is usage, and the *Utilization Fee* is charged when this threshold is exceeded. Finally, *Prob(Cancellation)* represents the probability that the loan will be canceled before maturity, and the *Cancellation Fee* is the cost associated with such an early termination.

Independent Variables	
CO_Firms	The natural log of one plus the number of the firm's customers that are co-owned with the supplier firm.
CO_Firms_MV	The natural log of one plus the market value of the firm's customers that are co-owned with the supplier firm.
COwn_#	The natural log of one plus the number of institutional investors who are shareholders of both the supplier firm and its major customers.

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Independent Variables	
Avg_Investors	The number of vertical common institutional investors scaled by the number of total institutional investors in the firm.
CO_AvgFirms	The number of the firm's customers that are co-owned with the supplier firm, scaled by the total number of customers that the firm has.
CO_AvgFirms_MV	The market value of the firm's customers that are co-owned with the supplier firm, scaled by the total market value of customers that the firm has.
COwn_D	A dummy variable that takes a value of one if the supplier firm has any common ownership with its major customers, and zero otherwise.
COwn_Avg_EW	The arithmetic average of the supplier firm *i*'s shares and customer *j*'s shares held by common investors. It is aggregated at the firm level by using an equal-weighted average across all its customers.
COwn_Avg_VW	The arithmetic average of the supplier firm *i*'s shares and customer *j*'s shares held by common investors. It is aggregated at the firm level by using a market value-weighted average across all its customers.
COwn_Avg_SW	The arithmetic average of the supplier firm *i*'s shares and customer *j*'s shares held by common investors. It is aggregated at the firm level by using a sales-weighted average across all its customers.
COwn_Value_EW	The total market value held by all the common investors of the supplier firm *i* and its customer *j*, scaled by the total market capitalization of the supplier *i* and customer *j*. It is aggregated at the supplier firm level by using an equal-weighted average across all of its customers.
COwn_Value_VW	The total market value held by all the common investors of the supplier firm *i* and its customer *j*, scaled by the total market capitalization of the supplier *i* and customer *j*. It is aggregated at the firm level by using a market value-weighted average across all its customers.
COwn_Value_SW	The total market value held by all the common investors of the supplier firm *i* and its customer *j*, scaled by the total market capitalization of the supplier *i* and customer *j*. It is aggregated at the firm level by using a sales-weighted average across all its customers.
HCO_Firms	The natural log of one plus the number of the firm's industry peers that are co-owned with the firm.
HCO_Firms_MV	The natural log of one plus the market value of the firm's industry peers that are co-owned with the firm.
HCOwn_#	The natural log of one plus the number of institutional investors who are block holders of both the firm and its industry peers.
HAvg_Investors	The number of horizontal common institutional investors scaled by the number of total institutional investors in the firm.
INST	The percentage of shares held by institutional investors
COwn_Pair	A dummy variable equals one if a supplier-customer dyad has any common ownership, otherwise zero.
Sum_COwn_Pair	The natural log value of one plus the number of common institutional owners for a supplier-customer dyad.
COwn_Value_Pair	The percentage of market value held by common institutional investors for a supplier-customer dyad, which equals to the total market value of shares for which the supplier-customer dyad is commonly held, then divided by the total market value of the supplier firm and customer firm.

Control Variables	
SIZE	The natural log of the total assets.
LEV	The ratio of long-term debt plus debt in current liabilities to total assets.
MTB	The ratio of the market value of equity to the book value of equity.
ROA	The income before extraordinary items divided by total assets.
Tangibility	Property, plant and equipment divided by total assets.
CFO_VOL	Cash flow volatility, which is the standard deviation of earnings before extraordinary items (scaled by total assets) over the past three years.
Z_Score	Altman *Z*-Score, equals to (1.2 × working capital +1.4 × retained earnings +3.3 × earnings before interest and taxes +0.999 × sales)/lagged total assets.
Rating	Bond rating dummy, which equals one if the firm has an S&P bond rating, otherwise zero.
Performance	Performance pricing dummy, which equals one if the loan facility uses performance pricing and zero otherwise.
LoanPurpose	Loan purpose fixed effects, including corporate purpose, working capital, debt repayment, acquisition, backup line for commercial paper, and others.
LoanType	Loan type fixed effects, including term loan, revolver, and 364-day facility.

Cross-Sectional Variables	
Redeployability	The asset redeployability measured by the BEA capital flow table.
Entry	The proportion of new firms to total firms within an industry, where the new firm is just established for one year or two years
AR	The accounts receivable of the supplier divided by lagged total assets.
Opacity	The prior three years' moving sum of the absolute value of discretionary accruals.
Duration	Common ownership duration counted in months of common institutional investors. For firms with multiple common owners, *Duration* equals the median value of investment lengths for all the common owners.

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